

Specialization Overview: *Becoming AI Fluent – From Curiosity to Capability*

This specialization guides learners through the evolving landscape of generative AI, starting from the basics of what AI is and how it works, and progressing toward fluency in using AI tools across domains and decision-making contexts. Like learning a new language or mastering a musical instrument, developing AI fluency takes time, exploration, and intentional practice. Learners will engage with real-world examples, apply tools hands-on, and critically reflect on both the possibilities and limitations of this transformative technology.

Across the three courses, learners will:

- **Demystify** generative AI and its practical uses.
- **Experiment** with AI tools to improve their own workflows.
- **Customize** and refine AI outputs to suit real-world tasks.
- **Evaluate** the ethical, legal, and social implications of AI adoption.
- **Strategically position** themselves to lead or adapt within an AI-augmented world.

Whether learners are curious beginners or professionals looking to stay ahead, the specialization is designed to build foundational understanding, foster confident application, and prepare them to critically and creatively interact with AI in their personal and professional lives.

Throughout the course we'll use an underlying theme of ***AI-augmented personalized learning*** to explain concepts through applicable examples. We'll also weave in the topic of “vibe coding” to showcase AI-augmented programming in an approachable way for learners with little to no coding experience but also as a powerful force multiplier for learners with extensive programming experience. Throughout the course learners will gain confidence that, ***“AI won’t replace me because I’m using AI to make myself irreplaceable.”***

Course 1: *Becoming AI Literate*

Course Narrative Overview

This course is a guided, hands-on introduction to generative AI. It's designed for people who want to move beyond headlines and hype, and actually understand *what's going on* when they use tools like ChatGPT, Claude, or NotebookLM. Learners will gain a foundational understanding of how large language models and generative systems work not just how to use them, but why they behave the way they do.

We'll break down essential concepts like transformers, attention, and context windows at a conceptual level, with no math or code required. We'll also examine different modalities (text, image, audio), introduce prompt engineering through everyday examples, and help learners build a strong mental model of how these systems generate, predict, and hallucinate. This course emphasizes building intuition: by the end, learners will know enough to use GenAI tools thoughtfully and will be prepared to go deeper.

Module Narratives

Module 1: Talking to a Machine like a Human Treat the AI like a stranger on a bus: curious, conversational, and open-ended. We introduce language models, how they generate responses, and what context windows mean. Through playful interactions and guided practice, learners develop baseline comfort while forming an intuitive understanding of how LLMs work.

In Module 1, we flip the traditional script. Instead of starting with definitions or theory, we begin by engaging directly with generative AI inviting learners to interact with these tools the same way they might talk to a curious, confident stranger on the bus. This metaphor sets the tone: be open, be playful, but don't overshare. Learners are encouraged to experiment with text, image, and audio generation tools using whatever is on their mind from everyday tasks to creative whims.

Once they've explored freely, we pause and introduce foundational concepts: how language models generate responses, what prompts are, and how thoughtful prompting can improve results. We also situate generative AI in the broader arc of artificial

intelligence providing a historical lens and a conceptual map of where LLMs fit into the bigger picture.

Armed with this new understanding, learners return to the tools they explored earlier this time applying best practices to craft more effective prompts. This module emphasizes iteration, intuition-building, and personal relevance. Learners will also begin reflecting on what they hope to gain from the course, setting personalized goals through an AI-guided reflection exercise. By the end of Module 1, they won't just be *using* GenAI they'll be thinking more critically and confidently about what it is and how it can serve them.

Learning Goals:

- Develop a baseline comfort interacting with generative AI tools across modalities (text, image, audio).
- Begin forming an intuitive mental model of how large language models (LLMs) work.
- Understand where generative AI fits within the broader history and scope of artificial intelligence.
- Learn and apply basic prompting strategies to improve AI interactions.
- Reflect on personal goals and consider how AI might support their day-to-day life or work.

Flow & Activities

1. Welcome and Orientation Video: “Talk to It Like a Stranger”

We open the module with a short, personable video that introduces the idea of treating GenAI like a confident stranger on the bus. Learners are reassured that no technical background is needed and are encouraged to engage with curiosity but also caution, avoiding personal or sensitive information. The metaphor sets up safe, exploratory engagement, and normalizes experimentation.

2. First Exploration: Prompt the Machine

Learners are asked to interact with a generative AI tool (e.g., ChatGPT, Gemini, Perplexity, or Coursera’s built-in chatbot). They are given a menu of prompt ideas to choose from or are encouraged to come up with their own. Examples include:

- A recipe for picky eaters
- A fun date night idea
- A logo for a pickleball club
- A simple app idea (e.g., stock tracker)
- An email to a colleague they’ve been avoiding writing
- Socratic Tutor Prompting Project (Personalized Learning):
 - Activity: “Hi, I’m a student studying [your subject]. You’ll act as my Socratic tutor using the module’s readings as source material ask guiding questions, don’t give answers.” Why it fits: Immediately gives learners a concrete “personalized learning” use case.

This activity is positioned as “just try it” there’s no expectation of mastery, only exploration.

3. Reading: A Gentle Introduction to Generative AI and LLMs

Next, learners read an accessible article that explains what generative AI is, what large language models are, and why they’re significant. The reading walks through key ideas like prediction, probability, and pattern learning, introducing terms like “transformer,” “context window,” and “attention” in simple, conceptual terms. This helps learners begin forming a mental model of what’s happening under the hood.

4. Video: “What’s Under the Hood?” (Lightboard Explainer)

A companion video reinforces the reading through visual metaphors and engaging explanations. It covers the idea that LLMs are more than just next-word predictors, that they rely on probability and attention, and that much of their power stems from emergent behaviors. The goal here is to demystify LLMs without overwhelming learners with math or code. Fine-tuning, back propagation, and training can be hinted at here but covered more in the Part 2, also part 2 is where we'll talk about vibe coding and LLMs as the final layer of abstraction in coding.

5. Prompting Basics Mini-Lesson

Before learners dive back into practice, they're introduced to foundational prompting strategies. A brief mini-lesson or infographic explains best practices like:

- Being clear and specific
- Providing relevant context
- Assigning roles to the AI (e.g., "act as a nutritionist...")
- Giving examples or constraints

This lesson ends with a quick "check your intuition" quiz where learners choose between prompts and predict which will likely produce better results.

6. Re-Prompting Practice: Now Try It With Strategy

Now that learners have a few techniques, they return to their earlier interactions and re-engage with AI this time with intention. If they asked for a recipe earlier, they now prompt with more context: "Act as a pediatric nutritionist. My kids hate vegetables but love cheese. Suggest three creative dinner ideas." They are encouraged to notice how the output changes. Enhance the socratic tutor.

7. Reading: The Bigger Picture Where GenAI Fits in the History of AI

This reading provides essential historical and conceptual context. It briefly surveys the evolution of artificial intelligence symbolic AI, expert systems, neural networks, deep learning and shows how generative AI fits into that lineage. A diagram helps learners visualize how GenAI is one piece of the much broader AI ecosystem.

Relate to other parts of the course – look at alphago or whatever and then say how we'll look at those parts.

8. AI-Guided Reflection: What Are You Hoping to Get Out of This Course?

Learners engage in a reflective dialogue with the course-specific Coursera chatbot, which is equipped with knowledge of the course structure. Through the interaction, they answer two guiding questions:

“What are you hoping to get out of this course?”

“How do you think GenAI might help in your day-to-day work or life?”

The chatbot suggests relevant modules or future modules that might align with the learner’s goals, helping personalize their learning journey from the start. Give an example prompt about personalized learning, “I want to use AI-augmented assistants to help me learn.”

Module 2: Choosing the Right Tool for the Task Not all AI tools are the same. This module compares general and domain-specific tools and introduces conceptual differences between models. We cover high-level ideas like embeddings and retrieval-augmented generation, grounding them in personal use. Learners start to see how to pick the right model for the job. Different tasks require different models, and this module is about building critical thinking skills to match the tool to the job. Whether working with text, images, audio, or code, learners will explore how underlying architectures (like GANs, VAEs, transformers, and diffusion models) shape the kind of content a tool can generate and why it matters.

We open with an exploration of how different tools handle similar tasks, inviting learners to compare their outputs and reflect on their strengths and weaknesses. Through readings and visual explainers, they'll develop a conceptual understanding of various GenAI architectures enough to build intuition without diving into math or code. They'll also hear from practitioners who regularly work with a range of tools, offering real-world context for how model choice impacts workflows and creativity.

This module's hands-on activity uses NotebookLM to demonstrate how retrieval-augmented generation (RAG) can personalize and reshape outputs. Learners will use it to generate a custom podcast episode from course materials and reflect on how well it aligned with their needs surfacing early conversations about AI's limitations and prompting them to consider where human oversight is still essential.

- **Learning Goals:**
Understand how different types of generative AI models (e.g., transformers, GANs, VAEs, diffusion models) work at a conceptual level and what types of outputs they produce.
- Evaluate which generative AI tools are most appropriate for specific tasks or modalities.
- Explore how RAG systems like NotebookLM use embedded knowledge to customize responses and reflect on the implications.

Flow & Activities

1. Video: “Which tool?”

We begin with a high level look at the world of generative AI beyond the chatbot. The goal is to illustrate the diverse ecosystem of tools and normalize tool-switching based on task context. Also explore how professionals navigate tool choice in the real world. What do they consider when choosing between models? How do they factor in task requirements, model reliability, and time constraints? This segment reinforces the theme of strategic use.

2. Reading: “Which Model for Which Job?”

A conceptual reading introduces the main families of GenAI models (transformers, GANs, VAEs, diffusion models), explaining what each is designed to do and what kinds of outputs they typically produce. It also touches on retrieval-augmented generation (RAG) and embedded knowledge. Visual diagrams will support learners in distinguishing between these architectures at a high level.

3. Lightboard Explainer Video: “What’s Under the Hood, Part 2”

This video builds on the previous module’s foundation, showing how different generative architectures function conceptually. Using analogies and animations, it helps learners see why a GAN excels at image realism, or how a VAE learns latent representations. The focus remains conceptual and intuitive, bridging into the more technical depth introduced in Course 2. Fine-tuning, back propagation, and training go into conceptual depth. Also introduce coding with LLMs as the final layer of abstraction for programmers.

4. Practice: Same Theme, Different Modalities

Learners pick a common theme (e.g., “curiosity” or “remote work” or “AI-augmented personalized learning”) and use multiple GenAI tools to generate different types of outputs:

- Write a short story using a language model
- Create an image using a text-to-image tool with that same story as the prompt
 - Feed the story into chatGPT and have it create a prompt for the imagen
- Generate a voiceover or song using an audio model
 - Again use chatGPT or built in coursera model to create a prompt for audio
- Generate creative image flash cards to help me study a topic on ____
- Optional: create a short video or animation

This allows learners to compare tool strengths and reflect on which model worked best for which modality and why. A comparison would be nice especially if we could find openly available tools that use different approaches. Maybe stable diffusion and chatGPT?

5. Activity: Build and Analyze Your Own AI-Powered Podcast. Short video on RAG and to introduce the activity.

Take the idea of embeddings from what's under the hood and expand to the macro level. This introduces RAG, not as a different modality but as a way to apply a principle to a different idea. In a Retrieval-Augmented Generation (RAG) system, tokenization breaks down text into smaller units (tokens), while embedding converts these tokens into numerical vectors that capture their semantic meaning. Tokenization is a preprocessing step, mapping text to integers, whereas embedding is a subsequent process that generates high-dimensional vectors used by the model to understand and find semantically similar text.

Learners upload a course-related document (e.g., the full narrative outline of the specialization or selected readings) into NotebookLM and use it to generate a personalized podcast or audio summary. They then reflect on:

- How well the tool summarized the material
- What was included or omitted
- Whether the output aligned with their intent

This introduces them to RAG-style tools and surfaces an early understanding of AI's limits and biases setting up Module 4's deeper dive into hallucinations and red flags. Compare this RAG-style overview to just asking the chatGPT for a summary of a "a cousera course on generative AI, I'm specifically interested in how it would help me as I'm a ___ focused on improving my skills in ___ and _____. (For non-native speakers could generate in their own language).

6. Reflection & Discussion Prompt: "What Did You Notice?"

In the discussion forum, learners share their observations from the podcast activity:

"What did your AI-generated podcast emphasize or skip that surprised you? What does that tell you about how these tools 'understand' content?" This helps learners begin noticing non deterministic behavior and opens the door to critical AI literacy.

7. Assessment: Applied Tool Matching Quiz

A light-touch quiz presents a few real-world scenarios (e.g., “I want to create a custom logo for a volunteer group” or “I need to summarize a dense scientific article”) and asks learners to match the scenario with the most appropriate GenAI tool or model type. Learners justify their choices, reinforcing applied understanding.

Module 3: Prompt Like a Pro (Without Being a Programmer)

In Module 3, learners level up their interactions with generative AI by mastering the fundamentals of prompt engineering and taking their first steps into context engineering (Andrej Karpathy). Prompting isn't just about asking better questions it's about guiding the model's focus and using its architecture to your advantage. This module emphasizes that prompting is a strategic skill rooted in the attention mechanism and shaped by the limits of the context window. Related to the context window and attention is 'tokenization' and how content is broken up before it's passed to the model.

We begin by helping learners understand what a context window is and how models "pay attention" to certain parts of an input more than others. We then move beyond basic prompt tips and dive into prompting as system design how clarity, specificity, and structure affect model behavior. Learners will explore tips for role setting, giving examples, and sequencing information, and they'll see that the model's full conversational history matters just as much as the last line of input.

To reinforce these concepts, learners will interact with GenAI tools in a "prompt-and-refine" loop starting with natural, unstructured input and working iteratively to develop stronger prompts. They'll also reflect on how context accumulates in a conversation and learn strategies to manage or reset it. A fun "Prompt Jeopardy" assessment gamifies this practice and reinforces strategic thinking around structure and intention.

By the end of the module, learners won't just know how to write better prompts they'll understand the architecture behind those improvements, and how to use context to shape outcomes in a more consistent, powerful way.

Learning Goals:

- Understand how the structure of a prompt through token interacts with model attention and the context window.
- Apply foundational prompt engineering techniques to improve the clarity, relevance, and control of AI outputs.
- Recognize the broader role of context engineering in shaping model behavior over the course of a conversation.

Flow & Activities

1. Reading: "Why Context Matters"

We open with a reading that introduces the context window the limited space a model uses to “remember” and interpret input. Learners are introduced to ideas like token limits, conversational history, and the importance of where information is placed within a prompt. The reading also frames prompting as more than a one-shot task: it’s part of a larger interaction sequence. Visual diagrams will help learners understand the flow of input through attention layers.

2. Video: “What’s in the engine”

This visual explainer builds-on the reading by showing how attention mechanisms operate within transformer models. Using analogies (e.g., spotlight or memory focus), the video helps learners see how some parts of a prompt are weighted more heavily than others and how this shapes the model’s response. The video also reinforces the idea that context can be actively designed through structure and sequencing.

3. Best Practices Guide: Prompt Engineering + Context Engineering

Learners receive a visual handout or mini-guide with essential tips for crafting strong prompts. This goes beyond “write clearly” to include:

- Assigning roles to the AI (“act as a travel planner...”)
- Personalized learning – useful prompts and materials to give a socratic tutor bot. Make your own tutor bot.
- Providing examples or constraints
- Sequencing information intentionally
Managing or refreshing context
- Also prompts for multi-modal content. Prompts could even include seeding the video with an image first instead of asking it to come up with it on it’s own. Or an audio track and asking it to continue it or asking for a detailed prompt from chatGPT to create an audio track in a different generation.
- BONUS: Using the current chat to summarize itself before starting a new session (when behavior degrades)

We introduce the Andrej Karpathy idea that prompt engineering is the tip of the iceberg, and that true fluency involves managing the full context of a conversation. We compare this to the “stranger on the bus” metaphor from Module 1 reminding learners that what you said earlier still influences what happens now...at least for a while.

4. Hands-On Activity: Prompting and Re-Prompting

Learners are given a prompt task (e.g., “help me draft a team meeting agenda,” or “generate a workout plan”) and start by describing it in their own natural language. Then, using the coursera chatbot:

- They ask the model to write a refined prompt based on their initial input.
- They use the refined prompt and note how the response improves.
- Optional: They ask the AI to adapt that prompt for a different model (e.g., a model with longer context windows or image capabilities).

They test variations using the same context in different ways and reflect on how the changes affected the results.

Hands-On Activity:

Prompt for a socratic tutor bot using specific materials, how about the prompt guide. So the bot acts as a socratic tutor on the prompt guide. Start this exercise with the tip to use the LLM to create it's own prompt to act a certain way just using natural language.

5. Discussion Prompt: “What Did You Learn About Prompting?”

Learners share one before-and-after example from the exercise and reflect on how prompting choices changed the outcome. Sample prompt:

“How did your second attempt compare to the first? What did you change and what happened?”

This reflection helps reinforce the link between prompt design and model behavior, and lets learners compare techniques with one another.

6. Assessment: Prompt Jeopardy!

In this light but strategic assessment, learners are given a series of goals and must choose or design prompts that would best achieve the outcome. For example:

- You want concise bullet points for a marketing email.
- You want the AI to summarize a dense legal document.
- You want an empathetic response to a customer complaint.

- Prompts that give tutor style outputs.

Learners select from several prompt options or construct their own. This reinforces the application of prompting principles in diverse, real-world tasks.

Module 4: Know the Limits, Spot the Bias Every model has limits.

By Module 4, learners have explored the power of generative AI and begun to master the art of prompting. Now it's time to zoom out, to look critically at what these systems get wrong, why it happens, and what it means. This module focuses on trust, transparency, and responsibility, introducing learners to some of the most common and consequential limitations of GenAI tools: hallucinations, bias, training cutoffs, and context limits.

We frame this exploration through the lens of the "Three R's":

- Responsibility – Understanding one's role in using AI ethically and thoughtfully.
- Red Flags – Recognizing signs of bias, unreliability, and hallucination.
- Retrieval-Augmented Generation (RAG) – Learning how to ground models with external data to improve accuracy and reduce hallucinations. RAG is a specific technique but the idea of wrapping other techniques around each other is the valuable idea. This will then lead to more advanced techniques like deep research and internet use and probably also multi-modality and 'agents' in future lessons.

We also revisit the "stranger on the bus" metaphor, this time to introduce the unsettling implications of AI behavior. The AI may sound helpful and confident, but it's not just remembering what you said, it's using your behavior to model humans like you. That behavior is part of what fuels model improvement, which raises essential ethical questions: How is your interaction data being used? Who benefits? And who might be left out?

Through a combination of readings, real-world case studies, hands-on experiments, and thoughtful discussions, learners will develop a more critical lens for evaluating GenAI outputs, and for participating in the broader conversation about how these tools should be designed and deployed.

By the end of this module, learners will no longer see GenAI as a magic box, they'll understand the hidden risks, the design tradeoffs, and their own role in ensuring responsible use. module 4 marks a shift in mindset: from experimenting with AI to **understanding its limits and implications**, preparing learners to move into more intentional and ethical application in module 5.

Learning Goals:

- Identify common limitations of GenAI tools, including hallucinations, bias, and training cutoffs and understand how they arise.

- Develop strategies to recognize and respond to problematic or unreliable AI outputs.
- Reflect on ethical considerations and personal responsibility in AI use.

Flow & Activities

1. Reading: “Red Flags and Blind Spots”

We begin with a conceptual reading that introduces the core failure modes of generative AI:

- Hallucinations – The model fabricates plausible-sounding content that’s not grounded in reality. Relate this to probability. The best probable sounding output may not be correct.
- Bias – Outputs reflect patterns in biased training data, even when unintentional. Relate this to the ‘weights’ of the model.
- Training Cutoff – Models have incomplete knowledge past a certain date, which can introduce subtle errors or reinforce outdated information.
- Context Window Limitations – Important information may be dropped or misunderstood when conversations become too long. This was a common theme of the previous lesson.
- Environmental considerations – LLMs are pretty energy intensive to train, and also to interact with. It’s important to recognize that the training is where the vast majority of energy input is and interacting with the trained model is fairly ‘inexpensive’.

This reading helps learners see how these issues are rooted in architecture and training—not just misuse.

2. Video: “The Quirks of the Machine”

This video walks through real-world examples of GenAI failure. From counting letters in “strawberry” to misrepresenting historical facts, learners will see where even advanced models fall short. The video also connects these issues to ethical stakes: mistakes in a recipe are one thing mistakes in healthcare, hiring, or policing are another.

3. Sidebar: “A Brief History of Bias in AI”

A short, reading or video offers a historical lens tracing the evolution from early machine learning models to today’s GenAI systems. It includes cautionary tales:

- Microsoft's Tay chatbot that quickly became toxic
- Diversity "overcorrections" in image generators that erased historical accuracy.
- Biases that persist despite developers' best intentions

The goal is to show that these issues are not new, but they're now scaled up and easier to overlook.

4. Ethical Reflection: "You Are the Product?"

Learners revisit the stranger on the bus this time framed less playfully. They're introduced to the idea that if access to these tools is free, it's because their interactions are valuable. The AI isn't learning from your facts it's learning from your behavior, and that raises privacy and consent concerns. Learners are introduced to the concept of data monetization.

5. Interactive Practice: Spot the Bias, Fix the Flaw

- Learners are given a set of GenAI outputs and asked to:
- Identify if the output shows signs of hallucination, bias, or outdated info
- Propose strategies for fixing or mitigating the problem (e.g., rephrasing the prompt, using retrieval-based tools, providing better context)

They're also encouraged to generate their own example by intentionally designing a prompt that leads to hallucination or biased outputs.

6. Video: "Retrieval-Augmented Generation (RAG) to the Rescue?"

This video explains how RAG systems can reduce hallucinations by grounding outputs in specific, verifiable sources. NotebookLM is used as an example, along with other hybrid models that combine LLMs with external knowledge bases. This connects nicely to module 2's work with NotebookLM and helps learners distinguish between freeform generation and grounded output. The key takeaway with this video isn't that RAG is the savior, but rather how we can wrap technologies together to help alleviate some of the unintended behaviors. This will lead into more advanced concepts covered in later lessons in C2 and C3 like agents and advanced thinking models. A key component to get across here is to relate RAG with the context window of a regular LLM. When we upload content to an LLM without RAG that entire content probably goes at the start of the chat. So it would be like we told our stranger to read the entire book first and then we'll have a conversation VS. RAG where the stranger was allowed to make those color coded tabs of the book and then when we have an interaction it will always go through its tabs and give you specific content related to whatever you ask. Example of uploading course materials to create a 'study

guide' to a platform like NotebookLM and the limitations/considerations of doing that in terms of ownership.

7. Reflection Prompt: "What Would You Trust This Model With?"

Learners are asked to consider the boundaries of trust. For example:

"Would you trust this model to plan your vacation? Grade your paper? Screen job applicants? Why or why not?" This exercise pushes them to think critically about risk, context, and power dynamics in AI use. Could also bring up topics related to policy around legal issues – maybe bring in a policymaker like city council person Taiysha Adams.

Module 5: Reflect, Apply, and Evaluate the Evaluator

Module 5 brings everything together. Learners have played, practiced, and questioned and now it's time to apply. This module emphasizes deliberate, responsible use of generative AI in learners' own lives and work. Whether they're writing content, organizing information, creating visuals, or asking hard questions, the focus is now on intentionality: not just what GenAI can do, but when, why, and how to use it responsibly.

This final module is about synthesis, reflection, and strategic positioning. Learners have explored what GenAI is, how it works, and how to use it thoughtfully. Now, they turn inward to apply what they've learned to something meaningful to them, and outward to consider where they fit in a world increasingly shaped by AI.

We introduce a core message that has threaded quietly through the course but now moves center stage: The future isn't humans versus AI it's humans alongside AI. "AI won't replace me because I'm using AI to become irreplaceable." To succeed in this future, we must understand not just what AI can do, but what it can't. This module is about finding the edges of usefulness, of reliability, of responsibility—and learning to build human value into those spaces.

Learners will complete a mini-capstone project that ties together all the prior modules' work. Some learners will design and document their own project, using GenAI to support a personal or professional learning goal. Others may choose to follow a guided, example path focused on personalized learning our "default track." In both cases, the goal is to demonstrate intentional GenAI use, reflect on its effectiveness, and then take the meta-step of evaluating the AI evaluator itself.

This closing module invites learners to practice not just doing AI, but thinking with AI strategically, critically, and personally. It's a shift from literacy to fluency, and from curiosity to confident action.

This module also looks forward. We give learners a sneak peek at advanced topics like embeddings, fine-tuning, and evaluation not to overwhelm, but to inspire. We highlight how technology can be used to improve itself (via RAG, for instance), and how understanding limitations allows humans to find their unique role in the AI ecosystem.

By the end of this module, learners won't just be "AI literate", they'll be AI intentional, equipped to continue learning, critically evaluating, and adapting as these tools evolve.

Learning Goals

By the end of Module 5, learners will be able to:

- Apply GenAI to a personally meaningful task and reflect on its usefulness and limitations.
- Evaluate the quality and appropriateness of AI-generated outputs through critical analysis.
- Analyze feedback from an automated evaluator and assess its accuracy, insightfulness, and blind spots.
- Reflect on how humans can create value alongside AI and where they personally want to position themselves in that ecosystem.

Flow & Activities

1. Reading: “The Edge is Where We Belong”

This reading reframes the relationship between humans and GenAI. It introduces the idea that understanding AI’s boundaries isn’t just an academic exercise, it’s how we find our footing. Where AI ends, human creativity, judgment, and values begin. This sets the tone for learners to think of AI not as a replacement, but as a tool whose very limitations illuminate human strengths.

2. Video: “Humans Alongside AI”

A short, inspiring video highlights real-world examples of humans leveraging AI effectively, not by doing what AI does better, but by doing what only humans can do: asking better questions, applying ethical judgment, making context-aware decisions, and shaping the future of work. The message: Don’t compete with the machine. Partner with it wisely.

3. Mini-Capstone Project:

Learners complete a mini-capstone project to demonstrate their understanding of the topics covered so far. This is not a technical or design-heavy project. It is an opportunity to demonstrate intentional, reflective, and critical use of AI.

Track A: Your Own Goal

- Learners revisit a personal or professional goal from earlier in the course—or choose a new one.
- They describe the task, how GenAI helped them approach it, and reflect on the process using guided prompts.
- They submit their narrative and supporting artifacts (e.g., prompts, screenshots, summaries).
- Vibe code something...

Track B: Guided Example – Personalized Learning with GenAI

For learners who prefer a default path, they use GenAI to plan and support a personal learning journey (e.g., “I want to learn graphic design,” or “I want to study for a certification”).

- A structured worksheet walks them through prompt design, iteration, and reflection.
- This track is pre-configured to receive automated feedback from the AI grader.
- Both tracks emphasize not just the use of GenAI, but why, how, and with what consequences.

4. AI Evaluation – Then Critique the Evaluator

Learners receive feedback from an AI-based grader either on their own submission (Track A) or on the default example (Track B). They are then asked to critique the critique:

- Did the AI catch what mattered?
- Did it miss or hallucinate anything?
- How would you have evaluated this project differently?

This activity invites learners to peek behind the curtain and develop meta-literacy, the ability to assess the very systems that assess them. Will need to come up with a specific rubric for this capstone assignment. Make it approachable but rewarding and worthwhile.

5. Discussion Prompt: “Where Do You Fit Alongside AI?”

To close the course, learners reflect on the mini-capstone experience. The reflection is mainly focused on their experience of creating a mini-project and then receiving an AI-based evaluation of that project.

This prompt helps learners articulate a future-facing stance, reinforcing the course's message that understanding GenAI is not just about mastery it's about identity, strategy, and impact.

This final module will bring everything together, reinforcing the importance of using AI thoughtfully and ethically.

Course 2 Summary: GenAI as a Force Multiplier

How to control AI, shape its behavior, and turn it into a productive partner.

Course 2 builds on the literacy developed in Course 1 and turns it into practical capability. The focus is on how learners can guide AI effectively, engineer context for better outputs, and design multi step workflows that increase productivity and clarity. The goal is to help learners move from casual use to purposeful, structured engagement.

Learners begin by exploring how prompts influence the way AI generates responses. This includes how models react to structure, examples, roles, ordering, and constraints. Learners also experiment with settings such as temperature, top p, and top k, which shape creativity, precision, and tone. From there, the course expands into context engineering. This includes the construction of the environment the model uses to work. Learners further explore advanced retrieval augmented generation, embeddings, and external knowledge sources that improve accuracy and reliability.

The course then shifts from single prompts to complete workflows. Learners practice chaining prompts together. They learn how to turn a summary into an outline, an outline into a draft, and a draft into a polished result. They design workflows for professional tasks such as writing, analysis, teaching, marketing, and coding. Learners also experiment with the idea of "vibe-coding", which treats natural language as the interface for building programs or structured logic.

Learners also explore multimodal generation. This includes how the same AI principles produce images, audio, and other media. They study the basics of image models such as GANs, VAEs, and diffusion models. They also explore audio models such as WaveNet and MusicLM. This section highlights how text, image, and audio information interact inside a model.

The course concludes with an emphasis on evaluating AI outputs. Learners develop the practice of checking for clarity, grounding, relevance, and reasoning quality. They are introduced to the idea of reasoning or "thinking" models, which produce structured, step-by-step predictions. This prepares learners for the next stage, where structured reasoning evolves into agent like behavior.

By the end of Course 2, learners will be able to:

- Engineer prompts and context to shape AI behavior
- Use retrieval augmented generation for grounded and reliable outputs
- Build multi step workflows that support personal and professional goals
- Understand the core architectures behind image and audio generation
- Evaluate AI outputs with a critical and informed approach
- Recognize the early signs of structured reasoning in AI models

Course 2 prepares learners to transition from using AI tools to understanding the components of AI systems. This sets the foundation for Course 3.

Course 3 Summary: GenAI Fluent

How to evaluate models, design agentic systems, understand world models, and prepare for the AI-augmented future.

Course 3 brings together the ideas and skills developed throughout the specialization. Its purpose is to help learners think at a systems level. Learners move beyond individual prompts and workflows and begin evaluating models, combining tools, and understanding the direction of the field. The course prepares learners to engage with AI thoughtfully, strategically, and responsibly.

Learners begin by exploring the broader AI model ecosystem. They learn how to compare frontier models such as GPT, Claude, Gemini, and DeepSeek. They also learn about the rapidly growing world of open source and open weight models such as Llama, Mistral, Qwen, and Phi. This section is paired with hands-on model comparison using LM Arena, which allows learners to compare model performance directly. Learners practice judging quality, identifying strengths, and choosing the right model for a given task.

From there, the course turns to evaluation. Learners study how reinforcement learning from human feedback influences model behavior. They examine how evaluator models score or critique outputs and how modern systems use self-correction and feedback loops. These ideas serve as the bridge from structured reasoning to the next major topic: AI agents.

Agents represent an important shift in how AI systems work and how we work with them. Learners explore how agents plan, use tools, call external functions, retrieve information, and manage multistep tasks across time. They learn how memory and personalization add depth and continuity to AI behavior. The course covers single agent systems as well as multi agent coordination and introduces frameworks such as LangChain, CrewAI, and function calling interfaces.

Learners then study Large World Models. These models integrate text, images, audio, diagrams, and physical reasoning into richer internal structures. This helps learners understand why LWMs are considered an important step toward more general intelligence. Learners study how LWMs build consistent representations of scenes, objects, and events. They explore how these capabilities differ from traditional language models.

The course concludes with a forward-looking fluency plan. Learners reflect on their goals and develop an AI integration strategy for personal and professional contexts. They evaluate multiple models, build conceptual workflows, and articulate where AI adds value and where human judgment remains essential. The aim is to help learners move forward with clarity and confidence.

By the end of Course 3, learners will be able to:

- Compare and evaluate leading AI models, including open-source systems
- Understand how modern AI is trained, aligned, and refined
- Design conceptual agent systems that use tools, memory, and planning
- Explain the role of multimodal reasoning and Large World Models
- Understand the relationship between reasoning, agents, world models, and future AI development
- Develop a personal strategy for working effectively alongside AI

Course 3 transforms learners from capable users into fluent thinkers who understand how AI systems behave, how they evolve, and how to work with them responsibly.